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Passive-dynamic ankle-foot orthoses change post-stroke lower extremity constituent work profile

Jacob T. Skigen [a,*], Corey A. Koller [b,c], Darcy S. Reisman [b,d], Zahra N. McKee [b,c], Shay R. Pinhey [b,c], Jason M. Wilken [e], Elisa S. Arch [b,c]

[a] Department of Biomedical Engineering, University of Delaware, Newark, DE, USA
[b] Biomechanics and Movement Science Interdisciplinary Program, University of Delaware, Newark, DE, USA
[c] Department of Kinesiology and Applied Physiology, University of Delaware, Newark, DE, USA
[d] Department of Physical Therapy, University of Delaware, Newark, DE, USA
[e] Department of Physical Therapy and Rehabilitation Science, University of Iowa, Iowa City, IA, USA

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ABSTRACT

Stiffness-customized passive-dynamic ankle–foot orthoses (PD-AFOs) have been shown to reduce the mechanical cost of transport (COT) of individuals post-stroke. However, the mechanisms underlying this reduced COT are unknown. Therefore, this study aimed to identify the factors driving COT reduction with PD-AFO use for individuals post-stroke. Results showed that changes in limb work were strongly correlated to changes in COT with the PD-AFO compared to No AFO in the paretic (tau = 0.637, $p < 0.001$) and non-paretic (tau = 0.621, $p < 0.001$) limbs. There was also a strong correlation between changes in limb work and changes in COT compared to SOC AFO in the paretic (tau = 0.569, $p < 0.001$) and non-paretic (tau = 0.503, $p = 0.003$) limbs. Conversely, changes in stride length and changes in COT were not correlated. Changes in COT between No AFO and PD-AFO were moderately correlated to the number of constituents that performed less mechanical work for both the paretic (tau = −0.462, $p = 0.009$) and non-paretic (tau = −0.402, $p = 0.025$) limbs. Compared to walking with SOC AFOs, there was a moderate correlation between COT and the number of constituents in the paretic limb (tau = −0.458, $p = 0.011$) but not the non-paretic limb (tau = −0.247, $p = 0.173$). These findings indicate that PD-AFOs reduce COT primarily through small changes in work across many lower limb constituents. Understanding how COT reduction occurs can help optimize PD-AFO design and possibly other rehabilitation interventions for individuals post-stroke.

1. Introduction

For many individuals post-stroke, plantar flexor weakness on their affected side is persistent and a barrier to restoring efficient gait (Nadeau et al., 1999; Pak & Patten, 2008). This weakness results in an inability to control forward rotation of the lower leg during mid-stance (Dickstein, 2008; Nadeau et al., 1999; Olney & Richards, 1996), resulting in kinematic deviations such as excessive ankle dorsiflexion (Jonkers et al., 2003) or persistent knee hyperextension (Mulroy et al., 2003; Olney & Richards, 1996). Plantar flexor weakness also diminishes the ability to generate forward propulsion during push-off (Nadeau et al., 1999; Olney & Richards, 1996). These gait deficits result in an increased metabolic energy cost of walking post-stroke (Detrembleur et al., 2003; Farris et al., 2015; Franceschini et al., 2003; Stoquart et al., 2012), contributing to decreased mobility, community engagement (E. Mayo, et al., 1999; Vestling et al., 2003), and overall health outcomes (Kruk, n.d.; Mokdad et al., 2005; Robison et al., 2009). Thus, reducing the energy cost of walking is a major goal of post-stroke rehabilitation.

Ankle-foot orthoses (AFOs) are commonly prescribed to individuals post-stroke to assist with walking. Passive-dynamic ankle–foot orthoses (PD-AFOs) are a class of AFOs that can support weakened plantar flexors primarily via a spring-like bending stiffness (Arch & Reisman, 2016; Bregman et al., 2009; Condie, 2008; Faustini et al., 2008). Mechanical energy is stored when the PD-AFO is deformed during stance-phase dorsiflexion and then returned during push-off to aid forward progression (Faustini et al., 2008). Previous research has shown that different

* Corresponding author at: 100 Discovery Blvd, Rm 340, Newark, DE 19713, USA.
E-mail addresses: jtskigen@udel.edu (J.T. Skigen), ckoller93@gmail.com (C.A. Koller), dreisman@udel.edu (D.S. Reisman), mckeezn@udel.edu (Z.N. McKee), spinhey@udel.edu (S.R. Pinhey), Jason-wilken@uiowa.edu (J.M. Wilken), schranke@udel.edu (E.S. Arch).

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PD-AFO stiffnesses can positively affect patient gait, including reducing the energy cost of walking (Bregman et al., 2011; Ploeger et al., 2019; Waterval et al., 2019; Waterval et al., 2020b). Further, optimizing PD-AFO stiffness for non-stroke neuromuscular populations based on individual metabolic cost has been shown to outperform walking shod without an AFO or with a standard-of-care (SOC) AFO prescribed by a physician using traditional prescription methods (Waterval et al., 2017; Waterval et al., 2020a; Waterval et al., 2021). A recent study in individuals post-stroke provided complimentary results, showing PD-AFOs with stiffness individually tuned to address plantar flexor weakness reduced the net mechanical cost of transport (COT) for many individuals compared to walking with No AFO or with a SOC AFO (Skigen et al., 2024). Unlike metabolic energy cost—characterized by oxygen consumption— COT is calculated as the total mechanical work by the lower limbs, normalized by stride length and body mass (Ebrahimi et al., 2017). Although they measure different forms of energy expenditure, previous research has shown that mechanical work and metabolic energy cost are related in individuals post-stroke (Donelan et al., 2002; Farris et al., 2015; Stoquart et al., 2012; Voloshina et al., 2013).

While it is clear from these prior works that stiffness-customized PD-AFOs can reduce the energy cost of walking, the mechanisms underlying that reduction remain unclear. Previously, researchers have assessed the effects of varying AFO stiffness on ankle mechanics (Ploeger et al., 2019) and found a relationship between increased ankle power and decreased metabolic energy cost with an experimentally optimized PD-AFO for individuals with unilateral calf muscle weakness (Waterval et al., 2021); but the mechanisms underlying the cost reduction were not identified. It is important to understand the source of this COT reduction in order to harness those mechanisms in AFO device design and customization as well as in rehabilitation protocols. As such, this study's purpose was to identify the factors driving changes in mechanical COT walking with the PD-AFO compared to No AFO and SOC AFO use for individuals post-stroke. We hypothesized that decreased mechanical COT is driven more by reductions in limb work than increases in stride length. Additionally, we hypothesized that changes in mechanical COT are negatively correlated to the number of lower extremity constituents that decreased work with PD-AFO use. In other words, this hypothesis theorized that reductions in COT were driven by reduced work across multiple constituents rather than larger reductions in a few select constituents.

2. Methods

Existing data from 20 participants from a study that showed stiffness-customized PD-AFOs improved the net mechanical COT of walking for individuals post-stroke (Skigen et al., 2024) were used. This study was approved by the University of Delaware Institutional Review Board and informed consent was obtained for all participants. Individuals greater than six months post-stroke with plantar flexor weakness, as defined by at least a 0.15 N-m/kg deficit in paretic peak plantar flexor moment during stance compared to a scaled speed matched typical value, were invited to participate. Participants also had to have been prescribed an AFO by a physician and be able to achieve at least five degrees of ankle dorsiflexion. Exclusion criteria included neurologic conditions other than stroke, inability to walk outside the home prior to the stroke or to walk for two minutes without assistance from another person (assistive devices such as a cane or articulating AFO that provided no dorsiflexion resistance were allowed), ataxic gait, bilateral paresis as a result of one or more strokes, total joint replacement or other orthopedic problems in the lower limb or spine that limited walking ability, and insufficient cardiovascular health. If a participant wore a SOC AFO that provided resistance to motion in dorsiflexion and could not walk without the AFO, the participant was withdrawn from the study.

Each participant made three visits to the laboratory. During the first visit, baseline data were collected while the participant walked without any AFO (No AFO). First, participants completed a 10-meter walk test with No AFO to determine their self-selected walking speed (Plummer et al., 2007). Gait analysis data were then collected as participants walked on a split-belt instrumented treadmill (Bertec Corp., Columbus, OH) for a minimum of 15 s at the walking speed measured during the 10-meter walk test. Participants could have a light touch on the handrails, if needed, and wore a safety harness. Retroreflective markers were attached to the pelvis, legs, and feet to track six-degree-of-freedom motion (Holden et al., 1997). Kinetic and kinematic data were collected at 240 Hz and 1200 Hz, respectively, using the treadmill force plates and a 13-camera motion capture system (Qualisys, Goteborg, Sweden). The data were filtered at 6 Hz and 25 Hz, respectively, using a 4th order zero-lag Butterworth filter. If the participant could not walk without an orthosis, they completed the test while wearing their SOC AFO if it was a simple hinged AFO that did not provide any resistance to dorsiflexion motion.

Following the first visit, a PD-AFO (Fig. 1) with a customized stiffness based on their level of plantar flexor weakness during No AFO walking was manufactured for each participant (Skigen et al., 2024). Participants then returned to the lab on two more occasions, to collect data while wearing their SOC AFO and then while wearing their PD-AFO. The same data collection procedures used for the first visit were used for these visits. Participants were given time to acclimate to wearing the PD-AFO by walking around the lab until they felt comfortable.

Kinetic and kinematic data were analyzed in Visual 3D to calculate limb work and stride length in each condition (No AFO, SOC AFO, PD-AFO). The Constituent Lower Extremity Work (CLEW) approach was used to segment each participant's limbs into four parts (hip, knee, ankle, and distal foot) and then calculate the average positive and negative work, normalized by body mass, done by each part during stance and swing phases – excluding distal foot, which only included values during stance – in each condition. This provides 14 different work values (positive and negative hip, knee, and ankle work during stance and swing; and positive and negative distal foot work during stance), hereby referred to as constituents, for each limb. These constituents were then used to create a "work profile" for each participant and assess the contribution of each constituent to the total. Mechanical COT was then calculated per limb for each condition by summing the absolute value of all constituents and normalizing by stride length (Ebrahimi et al., 2017). Net COT was then calculated by summing the COT of both limbs.

Group means for each primary outcome variable were calculated in RStudio (Posit team (2023). RStudio: Integrated Development Environment for R. Posit Software, PBC, Boston, MA) by averaging across participants within each condition. Skillings Mack tests were used to compare the average COT across the three conditions for the paretic and non-paretic limbs. When significant main effects were found, Wilcoxon signed-rank tests were used for post-hoc pairwise comparisons, with a

Fig. 1. Stiffness-customized PD-AFO.

Holm-Bonferroni correction to account for multiple comparisons. Secondary analyses were conducted if needed to further explore significant findings. The relationships between the change in COT and the difference in limb work and stride length between conditions were assessed using Kendall's tau correlations for each participant's paretic and non-paretic limb. Kendall's tau was also used to assess the relationship between the number of constituents that reduced work and the change in COT for each limb.

3. Results

3.1. Overall COT changes

On a group level, the mechanical COT of participants' paretic limb decreased by 0.16 ± 0.27 J/kg/m (n = 19) while wearing the PD-AFO compared to No AFO and by 0.07 ± 0.22 J/kg/m (n = 18) compared to their SOC AFO. The COT for their non-paretic limb decreased by 0.30 ± 0.41 J/kg/m (n = 18) compared to No AFO and by 0.15 ± 0.22 J/kg/m (n = 18) compared to their SOC AFO. Interestingly, while there was a downward trend in paretic COT with the PD-AFO, there were no significant differences between conditions; however, there were significant differences in non-paretic COT with PD-AFO use compared to both the No AFO ($p < 0.01$) and SOC AFO ($p < 0.01$) (Fig. 2). Individual participant's change in COT, limb work, and stride length, and number of constituents that performed less work with the PD-AFO for all comparisons are provided in the supplementary materials (Supplementary Data Table 1 and Table 2). One participant with a hinged SOC AFO could not walk without an orthosis and therefore no data were collected for the No AFO condition. Due to a technical issue, mechanical COT could not be calculated for one participant's non-paretic limb for the No AFO condition. Note that differences between conditions were calculated using only the participants who had data from both relevant conditions, while averages in Fig. 2 are based on the total number of participants in that condition. The average paretic limb work and stride length for each condition were 0.85 ± 0.29 J/kg and 0.81 ± 0.23 m (No AFO); 0.71 ± 0.24 J/kg and 0.76 ± 0.25 m (SOC AFO); and 0.76 ± 0.28 J/kg and 0.83 ± 0.27 m (PD-AFO). For the non-paretic limb, the average limb work and stride length for each condition were 1.38 ± 0.52 J/kg and 0.84 ± 0.22 m (No AFO); 1.09 ± 0.26 J/kg and 0.76 ± 0.25 m (SOC AFO); and 1.12 ± 0.32 J/kg and 0.84 ± 0.26 m (PD-AFO). Individual participant's limb work and stride lengths for all conditions are provided in the supplementary materials (Supplementary Fig. 1 and Fig. 2).

3.2. Paretic vs. non-paretic limb COT changes

After finding significant differences in the non-paretic COT across brace conditions but not in the paretic COT, secondary analysis was performed to compare whether the direction of mechanical COT change (increase or decrease) in the paretic and non-paretic limbs aligned with the direction of change in net COT for each participant. As seen in Fig. 3, for most participants, increases or decreases in net COT were driven by corresponding changes in the COT of *both* the paretic and non-paretic limbs with the PD-AFO compared to either No AFO (n = 15) or SOC AFO (n = 10).

Of the 15 participants with corresponding changes in both paretic and non-paretic COT between the PD-AFO and No AFO conditions, 11 had a larger magnitude change in their non-paretic limb than paretic. For the other four participants, the larger change was always in the non-paretic limb. Of the 10 participants that had corresponding changes in paretic and non-paretic COT between the PD-AFO and SOC AFO conditions, four had a larger magnitude change in the non-paretic limb. In instances of opposing changes between the paretic and non-paretic limbs when compared to walking with their SOC AFO, a decrease in net COT was more likely to be driven by a decrease in the COT of the non-paretic limb than the paretic; and net increases were only caused by an increase in the paretic limb's COT.

Compared to walking with No AFO, the PD-AFO condition on average saw reductions in mechanical work in more than half of the constituents on both the paretic (8.57 ± 2.17) and non-paretic (8.67 ± 2.92) limbs (Fig. 4). Fewer constituents on the paretic (7.33 ± 2.52) and non-paretic (7.61 ± 2.64) limbs had reductions in mechanical work with the PD-AFO compared to the SOC AFO condition (Fig. 4).

3.3. Limb work vs. stride length

When comparing the effects of changes in limb work and stride length on COT, limb work was the driving factor in COT reductions. Change in limb work was strongly correlated to change in COT with the PD-AFO compared to No AFO in the paretic (tau = 0.637, $p < 0.001$) and non-paretic (tau = 0.621, $p < 0.001$) limbs, as well as the PD-AFO compared to SOC AFO in the paretic (tau = 0.569, $p < 0.001$) and non-paretic (tau = 0.503, $p = 0.003$) limbs (Fig. 5). Conversely, stride length did not correlate with the change in COT between walking with the PD-AFO and No AFO in the paretic (tau = −0.228, $p = 0.186$) and non-paretic (tau = −0.124, $p = 0.501$) limbs, nor between walking with

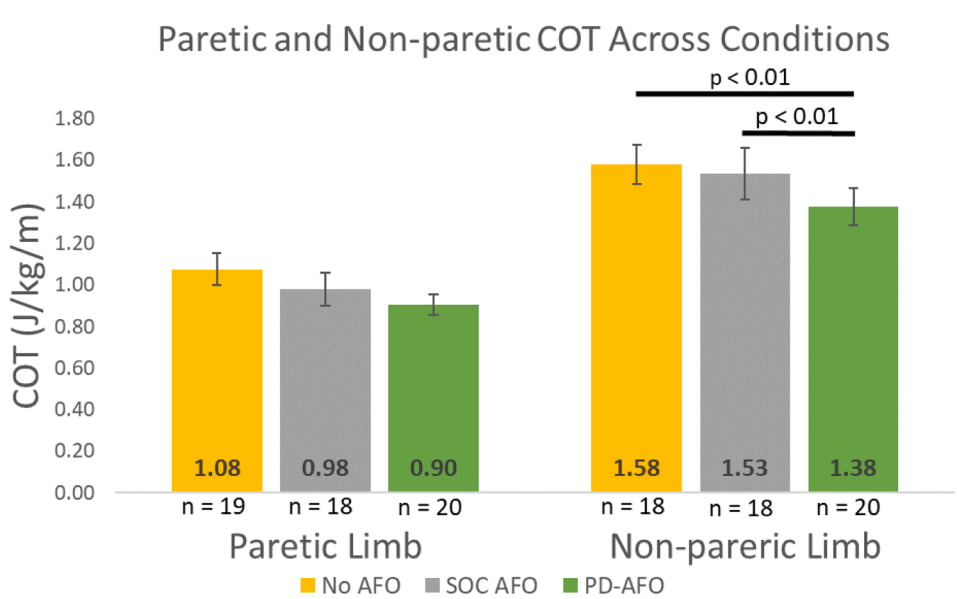

Fig. 2. Group means and standard error of mechanical cost of transport for paretic and non-paretic limb in each brace condition (J/kg/m).

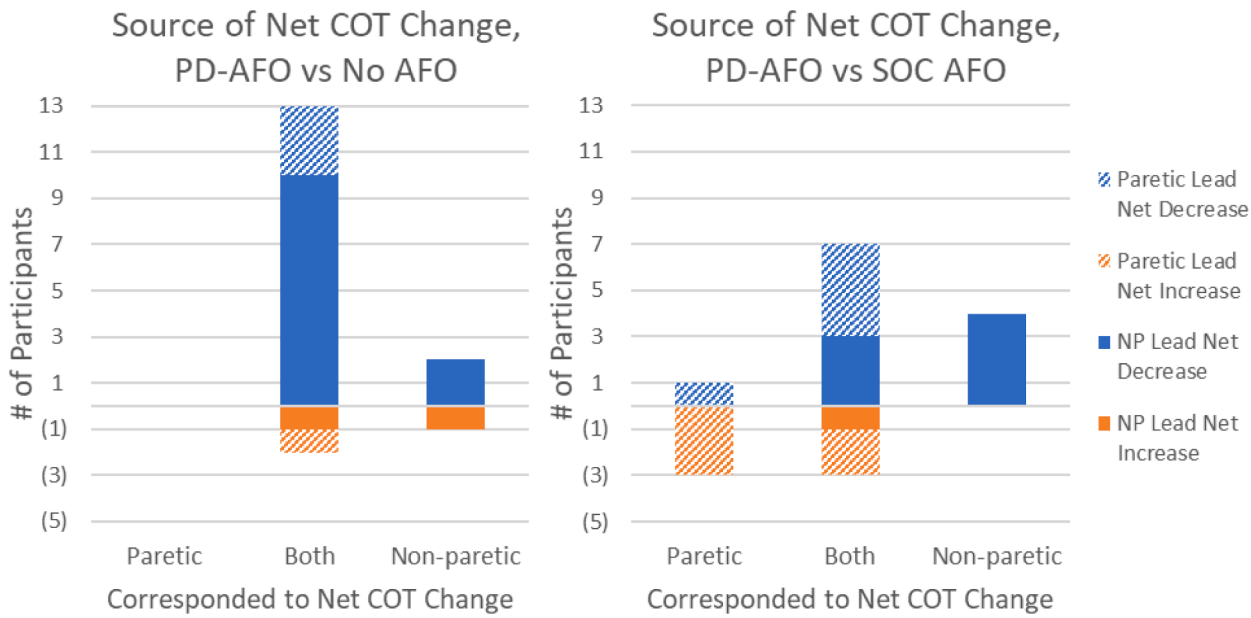

Fig. 3. Source of COT change by limb while walking with the PD-AFO compared to No AFO (left, n = 18) or SOC AFO (right, n = 18). Solid blocks represent participants who had the largest change in COT magnitude in their non-paretic limb. The participant with COT for only their paretic limb with No AFO and participants that did not walk in all three conditions were excluded.

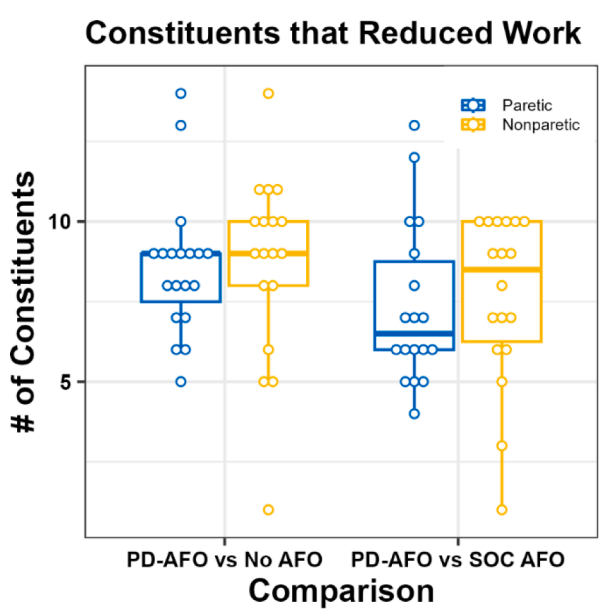

Fig. 4. Comparison of the number of constituents that reduced the amount of mechanical work done for each limb when walking with the PD-AFO compared to No AFO (left) and SOC AFO (right).

the PD-AFO and SOC AFO in the paretic (tau = −0.307, p = 0.081) and non-paretic (tau = −0.163, p = 0.369) limbs.

3.4. Number of constituents vs. change in COT

Comparing the number of constituents that decreased the work done with the PD-AFO to the overall change in limb COT revealed a negative correlation for both the No AFO and SOC AFO comparisons. When comparing PD-AFO to No AFO, there was a moderate negative correlation between the reduction in COT and the number of constituents that performed less work for both the paretic (tau = −0.462, p = 0.009) and non-paretic (tau = −0.402, p = 0.025) limbs (Fig. 6). Compared to walking with SOC AFO, there was a moderate correlation between the number of constituents that reduced the amount of mechanical work done with the PD-AFO and the decrease in COT in the paretic limb (tau = −0.458, p = 0.011) but not the non-paretic limb (tau = −0.247, p = 0.173). Notably, no two participants' work profiles changed the same way with the PD-AFO in terms of which constituents did less work and the magnitude of the reductions.

4. Discussion

This study aimed to identify the factors contributing to the reduction in COT while walking with a PD-AFO compared to No AFO and a SOC AFO for individuals post-stroke. The results showed the primary contributor of decreased COT was reduced mechanical work done by many lower limb constituents. Further, in most instances of net decrease in COT with the PD-AFO, the non-paretic limb played a larger role in that COT reduction than the paretic limb.

Investigating COT in the paretic versus non-paretic limbs revealed insightful findings. When comparing COT while walking with the PD-AFO to No AFO, the majority of participants had a larger change in their non-paretic limb than their paretic limb (Fig. 3). Previous research showed that for individuals post-stroke, paretic limb weakness necessitates non-paretic limb compensation to maintain walking speed (Mahon et al., 2015; Roelker et al., 2019; Stoquart et al., 2012). This reliance on the non-paretic limb usually represents an impairment of the inverted-pendulum model of walking observed in healthy individuals, which has been linked to increased net mechanical work in individuals post-stroke (Balbinot et al., 2020; Detrembleur et al., 2003). In that context, our results suggest that the PD-AFO reduced the level of non-paretic limb compensations, which produced a greater reduction in net COT than the direct enhancement of the paretic limb. Future research should investigate whether the reduction in non-paretic compensation with a PD-AFO correlates to closer adherence to the inverted-pendulum walking modality. Variability in participants' SOC AFOs made this SOC AFO to PD-AFO comparison more complex. However, while slightly more participants saw a larger change in their paretic than non-paretic limb, instances where the non-paretic limb had the larger change were more likely to cause a net COT decrease. This is in contrast with instances where the paretic limb had a larger change, which caused an increase in net COT as often as it caused a decrease.

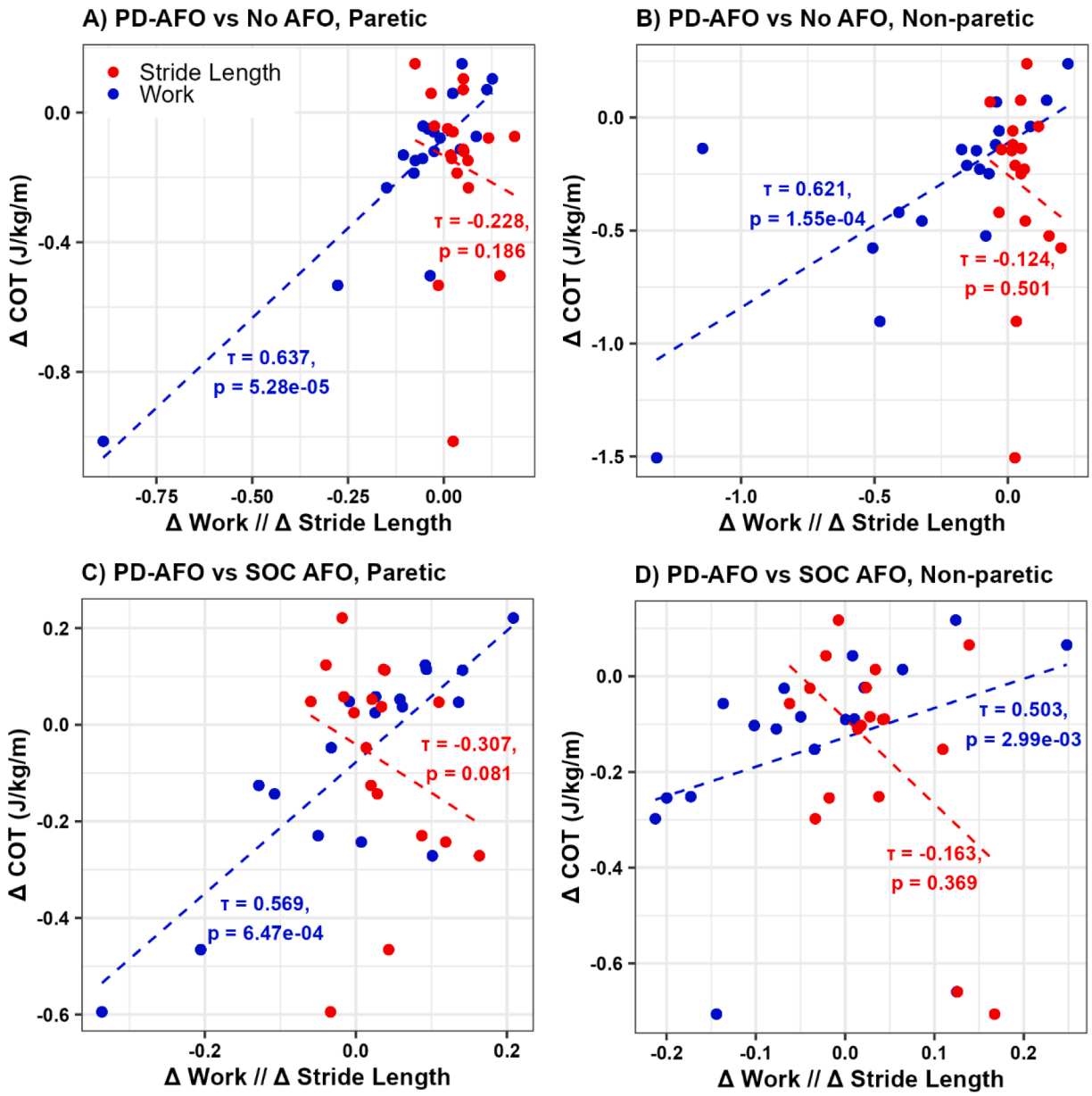

Fig. 5. Relationship between change in COT and change in both limb work and stride length when comparing: **A)** PD-AFO and No AFO walking in the paretic limb, **B)** PD-AFO and No AFO walking in the non-paretic limb, **C)** PD-AFO and SOC AFO walking in the paretic limb, **D)** PD-AFO and SOC AFO walking in the non-paretic limb.

Therefore, these results further support the conclusion that, energetically, PD-AFOs act by primarily reducing the compensatory load on the non-paretic limb. The non-paretic limb's role in reducing COT was further highlighted by the finding that during PD-AFO use, the non-paretic limb had, on average, more constituents with reduced work than the paretic limb (Fig. 3). In summary, results suggest that while stiffness-customized PD-AFOs can enhance the paretic limb, a larger benefit for many participants is lessening the burden on their non-paretic limb.

The hypothesis that COT decreases with PD-AFO use were driven primarily by reductions in limb work rather than increases in stride length was supported. These results align with previous studies that show PD-AFOs change the amount of work done by the lower limb system (Arch & Stanhope, 2015; Bregman et al., 2012; Ploeger et al., 2019). In contrast to limb work, there were relatively small changes in stride length across conditions for most participants. Even when the Kendall's tau correlations were run with outliers removed, conclusions remained the same (Supplementary Data Table 3). Notably, the minimal changes in stride length may be because data were collected on a treadmill while participants walked at the same speed across all conditions. Maintaining speed was necessary for comparisons across conditions, but it may have hidden changes in stride length that would have occurred with PD-AFO use in the free-living environment. Future research should investigate if larger changes in stride length occur with PD-AFO use when walking speed is not constrained, and if these potential changes impact COT.

The hypothesis that the number of constituents that reduced work would be negatively correlated with the change in COT was mostly supported, with all comparisons showing moderate correlations except for the non-paretic COT for the PD-AFO to SOC AFO comparison. These results indicate that the decrease in COT is due to a summation of multiple changes across the entire limb. However, the distribution of the magnitude of all work reductions can be seen in Supplementary Fig. 3. These findings align with prior research that showed that wearing a

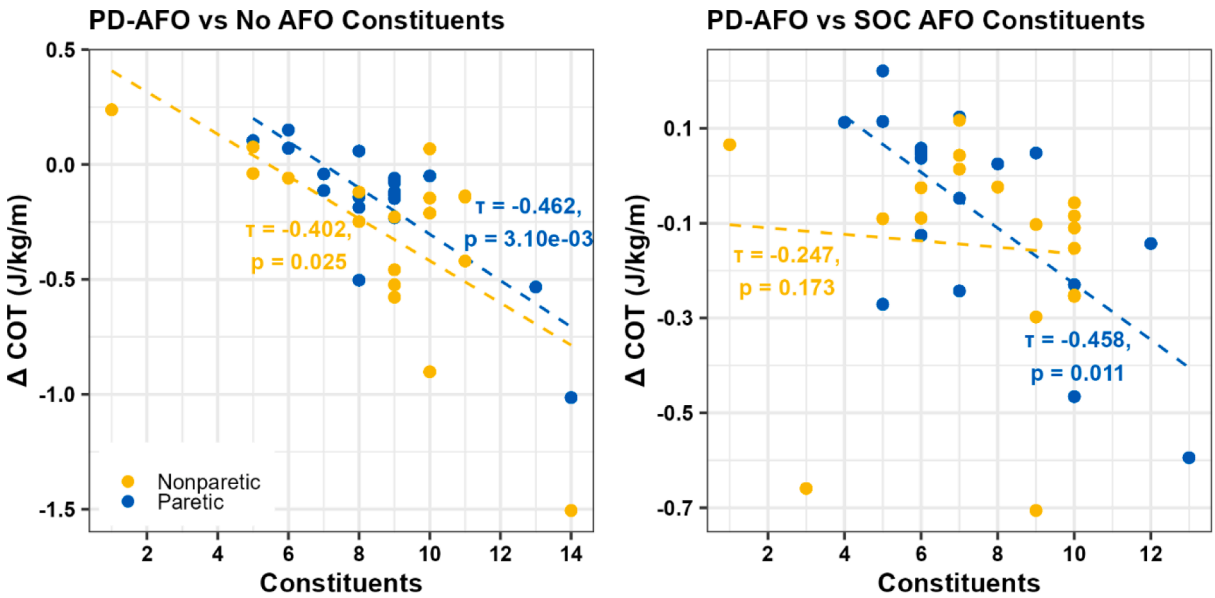

Fig. 6. Correlation between change in COT and number of constituents that decreased work magnitude when walking with the PD-AFO compared to No AFO (left) and SOC AFO (right).

passive-dynamic orthosis changed the work done throughout the entire limb, not just the ankle (Bregman et al., 2012), although that research only focused on the affected limb. Additionally, prior research has shown that plantar flexor weakness in individuals post-stroke often leads to a compensatory reliance on their hip flexors at push-off (Nadeau et al., 1999). Therefore, using a PD-AFO to augment weakened plantar flexors and reduce the need to rely on this and similar compensatory strategies, may allow for reorganization across participants' entire work profile to be more efficient. The disparate changes in each participant's work profile, in terms of which constituents reduced work and their magnitudes, suggests that every participant "self-organized" the changes in their own way; and this system-wide reorganization may be more instrumental to reducing COT than any specific constituent(s). Future studies should investigate whether individuals' self-organized changes are optimal for reducing COT or if providing guidance while walking with the PD-AFO can further improve their performance. Additionally, future work should investigate how best to harness this "self-optimization" when designing and customizing AFOs. The lack of significant relationship between the number of constituents that reduced mechanical work and the non-paretic COT change when comparing PD-AFO and SOC AFO walking may be due to participants already partially adjusting their non-paretic limb mechanics with their SOC AFO. As such, there would be less room for improvement with the stiffness-customized PD-AFO. This analysis further demonstrates the necessity of looking beyond the paretic ankle and considering the entire lower limb system when assessing the impact of AFOs on post-stroke gait.

Some limitations should be noted. First, PD-AFO data were collected the same day the participants were given the PD-AFO. Although participants were given time to acclimate to the orthosis, results may differ with longer-term PD-AFO use. Second, in these data the work from the AFO (SOC AFO or PD-AFO) cannot be distinguished from a participant's paretic ankle. However, the results showed that the main source of COT change when wearing the PD-AFO was not the paretic ankle; but rather, a series of changes across the entire lower limb system. Nevertheless, future studies should investigate how much of the work done by the paretic ankle is being generated by the AFO. Third, to our knowledge, there is no established minimum magnitude of change in work, for any application, that is considered clinically significant. As such, we deemed it more appropriate to include all reductions in our analysis instead of imposing an arbitrary threshold as to what qualified as a change in work. Defining a clinically meaningful threshold for changes in mechanical work is an important goal for future studies. Finally, our previous work showed that walking with a PD-AFO may result in a meaningfully faster walking speed compared to No AFO and SOC AFO (Skigen et al., 2024). Given that COT and other biomechanical variables are speed dependent, it was necessary to collect data with the participants walking at their baseline speed for all conditions. However, this may have limited the detection of speed-related changes.

Results from this study provide important new insights that can be used to advance rehabilitation interventions to further improve walking ability and outcomes for individuals post-stroke. The finding that there were small adjustments across the entire work profile with PD-AFO use motivates future research to investigate methods for leveraging this work profile reorganization to further enhance outcomes with PD-AFO use. Further, this study highlights the importance of considering the effect on the entire lower limb system – notably the non-paretic limb – when designing and customizing AFOs, and likely other rehabilitation interventions, for individuals post-stroke. Overall, this study is the first, to our knowledge, to reveal that PD-AFOs do not simply enhance post-stroke paretic ankle mechanics and instead have a complex influence on the entire lower extremity work profile – a finding that should drive future work.

5. Conclusion

In this study, the mechanisms through which stiffness-customized PD-AFOs improve the net mechanical COT for individuals post-stroke were examined. Findings indicated that COT reductions were the result of reduced mechanical work being performed by the lower limb system rather than increased stride length. More specifically, reductions in limb COT were achieved through a combination of multiple changes in work performed by many constituents across the lower limb system, with changes in the non-paretic limb constituents often being a larger driver of net COT change than the paretic limb constituents. This study revealed an important understanding of the contributions of constituents to COT reductions, which, to our knowledge, have never been examined before. With this foundational understanding, future research can work to harness these mechanisms to advance the design and prescription of AFOs as well as other rehabilitation interventions for individuals post-stroke.

CRediT authorship contribution statement

Jacob T. Skigen: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Investigation, Formal analysis, Data curation. **Corey A. Koller:** Writing – review & editing, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Darcy S. Reisman:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Zahra N. McKee:** Writing – review & editing, Investigation, Formal analysis. **Shay R. Pinhey:** Writing – review & editing, Investigation, Formal analysis. **Jason M. Wilken:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Elisa S. Arch:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbiomech.2024.112414.

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